ALP-ttbar Toy MC v1.6 → v1.7: From a Field and Material Scan to a Preregistered 3D Vertex Fit

Won Shik Paik (Independent Researcher, Auckland, New Zealand) · overview of three Zenodo records · 7 November 2026

Toy study motivated by Alimena et al., arXiv:2610.38932, which extends Rygaard et al., JHEP 10 (2023) 138

What this document contains. Page 1: this overview. Pages 2–4: v1.7 post-registration results (doi:10.5281/zenodo.24011489). Pages 5–8: v1.7 preregistration, frozen and public before any v1.7 result (doi:10.5281/zenodo.23995668). Pages 9–12: v1.6 post-registration results (doi:10.5281/zenodo.23952888). The three reports are reproduced unchanged.

Disclaimer. Independent personal project, not affiliated with the authors of the referenced papers or with the CMS, LHCb or ALICE Collaborations or the FCC study. Kinematics are parametrised and detector numbers assumed: only qualitative trends and ratios are meaningful; no claim is made about the performance of a real detector.

1. The question

For the lightest ALPs (17 MeV) the displaced e^+e^- pairs look almost like photon conversions in the tracker material. In this toy the conversion veto then costs a large factor in the expected limit ("limit degradation", $c\tau = 1$ mm). v1.5 introduced a two-hypothesis vertex fit (conversion vs. ALP opening angle) in the transverse view; it gave no gain at 17–50 MeV. v1.6 and v1.7 ask what could recover it.

2. v1.6: magnetic field and material budget (preregistered)

configuration	17 MeV	50 MeV	100 MeV	200 MeV
R: 3.8 T, nominal material (baseline)	127×	68×	28×	8.2×
B02: 0.2 T, nominal material	188×	58×	23×	10.9×
M05: 3.8 T, half material	65×	36×	16×	5.7×
M025: 3.8 T, quarter material	35×	20×	8.8×	2.1×

Material is the effective handle (127× → 65× → 35× at 17 MeV); lowering the field does not help the lightest ALPs. Four of seven predictions confirmed, three refuted. The v1.5 Wilks failure (boundary null) is recorded in an appendix.

3. v1.7: adding the r–z opening angle (preregistered)

In the r–z view a vertex shift cannot absorb a polar opening and bremsstrahlung barely changes the polar angle. v1.7 adds z hits to the unchanged v1.6 baseline and tests $\Delta\chi^2_{3D} = \Delta\chi^2_{T} + \Delta\chi^2_{L}$, with three z geometries.

selection (3.8 T, nominal material)	17 MeV	50 MeV	100 MeV	200 MeV
transverse only (= v1.6 R)	127×	68×	28×	8.2×
3D, pixel z (default)	127×	64×	24×	6.1×
3D + stereo strips	127×	61×	21×	4.5×
3D, z at every layer (all-pixel)	117×	38.5×	13.1×	3.9×

The r–z view separates ALP pairs from 50 MeV on, and z at every layer roughly halves the loss at 50–200 MeV. At 17 MeV it does not: few pairs have enough z hits, and electrons that lost energy to bremsstrahlung scatter more, which fakes polar openings in 7.4% of conversions. Six of seven predictions confirmed (P1 at the edge of its range); P7 (≥ 25% gain at 17 MeV with z at every layer) refuted: 7.8%. All validations passed; one sanity-gate failure was found and fixed, and is reported as a deviation.

4. Take-aways and next steps

• Across v1.6–v1.7, only reducing material reduced the 17 MeV loss substantially in this toy; field and r–z information helped heavier ALPs.
• Planned directions: a 3D Gaussian-sum fit; all-pixel trackers with reduced material, in the context of next-generation designs (ALICE 3, FCC-ee), without performance claims; a joint 3D fit; a dimuon control study.
• Every version follows the same order: preregistration frozen on GitHub and Zenodo → computation → post-registration report.

Records

record	DOI
v1.7 post-registration (pages 2–4)	https://doi.org/10.5281/zenodo.24011489
v1.7 preregistration (pages 5–8)	https://doi.org/10.5281/zenodo.23995668
v1.6 post-registration (pages 9–12)	https://doi.org/10.5281/zenodo.23952888
v1.6 preregistration	https://doi.org/10.5281/zenodo.23942338

Code, data products and READMEs: github.com/wspaik5 (each Zenodo record links its repository).